

Improving Information Security Risk Management in Resource-Limited Universities using OCTAVE Allegro with Data Recovery Mechanism

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Abstract—Educational places like universities are getting more and more into using databases and computer systems for handling student info and staff stuff, plus all the teaching services. It is basically a must now for running the academic and admin sides of things at school[9], [11].

But with all that reliance on tech, there are bigger risks popping up, like losing data or getting hit by cyberattacks, or even services just shutting down when you need them. I think that is the main worry here [10], [7].

This paper is trying to help out universities that do not have much money or tech support, showing them how to handle these security risks better. It uses this method called OCTAVE Allegro, which looks at the risks themselves, plus ways to recover data and manage disasters if they happen. The example they use is Sanaa University, which struggles with not having enough resources or good technology. Kind of like a lot of places in similar spots [2], [3].

The goal is to figure out ways to boost info security for schools facing these problems all the time, even if they are short on funds. It covers what data needs protecting, spotting the threats, and checking if they could really happen or if they would mess up the systems bad enough. To sort that out, there is this risk assessment matrix thing, which basically weighs how likely a threat is and what kind of damage it might do.

Then it suggests some practical steps to make data safer, such as doing backups regularly, having a reliable backup plan, copying over the key databases, and setting up a solid disaster recovery setup. And there are more ideas like that. Doing those could really improve how secure the information stays [6].

Merging the risk checking with the recovery plans seems like a smart way to keep things protected overall. For universities especially, this matters a ton because it lets them keep going smoothly even if something unexpected hits. Universities often lack the fancy tech resources

anyway, so being prepared means they can still handle classes and student needs if a system crashes [5], [8].

It feels like the whole point is just to have some kind of plan ready for when trouble comes. That way, schools can keep delivering what students need without big breaks in service. By getting ahead on risks and recovery, they cut down on the chaos from problems and keep their operations running. In the end, it builds up trust from students, which any university knows is super important, though sometimes it is hard to see how it all connects right away.

I Introduction

Universities these days rely a lot on digital stuff for student records and admin tasks. Information systems handle registrations, courses, finances, and all that. It just seems essential now for keeping things going [9], [11].

But yeah, with more tech comes more issues. Cyberattacks happen, unauthorized access occurs, data gets lost, or systems crash. This hits hard in schools, messing with classes and sensitive info like grades. Especially if the place doesn't have much budget or support to deal with it. I think that's when problems pile up faster [10], [7].

In underfunded spots, spotting risks is tough. They might skip proper security setups or threat plans. So needing a method to assess risks and find fixes makes sense. Something straightforward to check what's important and how to handle threats without wasting effort.

OCTAVE Allegro is one of those methods. It helps pick out key information assets, spot threats, and gauge the risks. You focus on what's critical first and decide protections based on that. For places with limited resources, prioritizing like this avoids spreading thin. It seems practical, lets you use what you have better, and cuts down vulnerabilities [2], [3], [5].

People often talk about risk management but forget the aftermath. Like recovering data after a breach or dealing with disasters. That's a gap, I suppose. Even strong prevention might not save you without a recovery plan. Combining both could strengthen things overall and keep operations smoother [8].

This study aims to propose integrating OCTAVE Allegro with data recovery and disaster planning. Using Sana'a University as an example. For budget-tight universities, this might improve security and keep education on track. Not sure if it's perfect, but it could work somehow. Things like that stand out as overlooked.

II Related Work

Numerous studies have illustrated the management of information security risks in different types of organizations. Experts develop methods and frameworks for identifying, assessing, and managing risks and vulnerabilities quite frequently within risk management. Educational institutions rely heavily on their information systems. Due to which they are often open to cyber threats that can disrupt classes or administration [6], [7] .

Many researchers refer to standards like the ISO 27005 Assessing Risk and the NIST SP 800-30 Guide for Conducting Risk Assessments to handle them. The authors outline stages to recognize essential information, detect threats, and estimate the odds of occurrence. After that, organizations will better protect data and reduce what happens if a breach takes place. The clearer picture of assets, threats, and decisions on risks gives us. Ultimately, this enhances overall protection and diminishes the impact of incidents, which, in these times, do seem like a big deal for businesses [6] .

The OCTAVE technique originates from the Software Engineering Institute at Carnegie Mellon University. This is to get insight into the key assets and threats that would be used to hit the organization. Research indicates that the effectiveness of OCTAVE Allegro is bolstered by its focus on the information itself rather than any technology such as computers or networks. That change in focus is logical. It digs deep into the stronghold. The Allegro version is more structured, which helps prioritize key assets and check threats and weaknesses against them. In all, OCTAVE appears to be useful for enhancing security [2], [3].

Some studies research information security in universities in particular. People say networks are complex with a multitude of users and systems to counter, which raises the chances of attacks or leaks. This implies the need for an all-encompassing strategy. Policies, technical protection, and education are important to get started on safe hazard practices [9].

Crafting disaster recovery plans is equally essential to maintain enterprise functionality should data or systems break down. According to the 3-2-1 rule, you need three copies of your data on two types of media with one copy offsite. Database replication is like having an extra copy of the database, so if one fails, the other takes over. By eliminating risks from ransomware and other nasty stuff, you're unlikely to suffer data loss or downtime. Being prepared with these essentials allows you to safeguard information and make systems operational [5], [8] .

Despite a lot of risk management and recovery work that exists in their own right, there isn't a great deal that tries to tie this together. Notably for schools that don't have the luxury of deep pockets and also have setup problems. This research is important as it combines the analysis of OCTAVE Allegro with concrete recovery ideas that improve security and allow smooth running of education. It seems like there's more to build on, though I'm not certain how it all fits together just yet.

III Problem Statement

Universities in modern times refer to information systems for running their various activities. Also, within an university setting, there are a variety of student data management, online registration system, grades management, financial and administrative systems, etc. The increasing dependence of stakeholders on the digital medium has made protection of information assets and continuity of the digital system more important [10].

Despite this, many universities struggle to manage information security risks, especially in low-tech, low-budget environments. The absence of plans for dealing with cyberattacks, data loss and system failure is visible in such settings [11].

Sana'a University is an example. Any failure in its information systems/data loss would affect vital processes such as registration, grading, and access to transcripts, among others. It can also harm the continuity of educational and administrative services apart from the risks of loss or unrecoverability of sensitive data.

Given the many frameworks and methodologies available for information security risk management, there are challenges in their application in universities. This mainly occurs when there are no clear mechanisms to integrate the risk analysis with data recovery and disaster recovery.

The research problem thus lies in the absence of an integrated framework that integrates information security risk analysis based on appropriate methodologies such as OCTAVE Allegro with data recovery and disaster recovery mechanisms considering the inadequate level of technology in the universities. This research would address this problem by proposing a model that is practical in nature and helps in the improvement of information security risk management for the continuity of educational system services [7], [8].

IV Research objectives

This research aims to provide an integrated framework for improving information security risk management in universities with limited technical and financial resources, while incorporating effective data recovery and disaster recovery mecha-

nisms. To achieve this goal, the research seeks to accomplish the following specific objectives:

- Identifying critical information assets at Sana'a University, including databases, electronic systems, and networks that support academic and administrative operations.
- Analyzing potential threats that could affect information systems, such as cyberattacks, unauthorized intrusions, data loss, and server outages.
- Assessing the risks associated with assets and threats using the OCTAVE Allegro framework to determine risk levels and treatment priorities.
- Designing an effective data recovery and disaster recovery mechanism to ensure the continuity of academic and administrative operations in the event of security incidents or technical failures.
- Proposing a model for improving information security risk management in universities facing technical resource challenges, which can be applied in similar environments to ensure data protection and the continuity of educational services.

V Methodology

This paper uses a case study approach [6]. I picked that because it fits for trying out the OCTAVE Allegro framework right at Sana'a University. The main goal here is to look into information security risks [2], [3]. And then, sort of build in some data recovery stuff to keep academic and administrative services running without big interruptions.

It seems straightforward, but the methodology breaks down into a few steps :

- Identify critical information assets:
 - Gather information about databases, electronic systems, and networks that support academic and administrative operations.

In a university setting, there are all these databases and electronic systems that keep things running for classes and admin stuff. Like, student records are stored in big databases, and networks

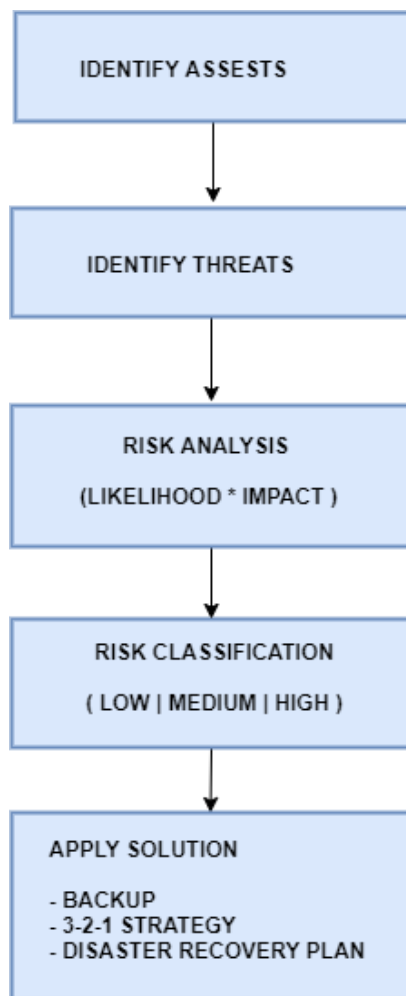


Fig. 1. Proposed methodology Steps based on OCTAVE Allegro [2], [3]

connect everything so professors can access materials or the finance office can handle payments. I guess the main idea is to figure out what supports academic operations, such as the learning management system where assignments get posted, and administrative ones like HR databases for employee info.

Gathering information about these seems tricky at first, but it involves looking at what holds sensitive data. Electronic systems include email servers and enrollment platforms that track who is taking what courses. Networks tie it all together, making sure data flows without issues. Some parts feel obvious, others

not so much.

- Identify the most important assets that require intensive protection.

The most important assets probably need the strongest protection because if they go down or get hacked, it messes up everything. For example, the central student database with personal details like SSNs or grades, that one stands out as critical. It requires intensive measures, like encryption and constant monitoring. Administrative networks for financial transactions also seem key, since money is involved and breaches could lead to real losses.

the academic side has assets like research databases that store intellectual property, those might need heavy safeguards too. Networks supporting classes, they protect a lot of operations indirectly. Its kind of messy to prioritize, but yeah, the ones with the most sensitive or vital data come first.

- Identify potential threats:

- Analyze cybersecurity risks such as cyberattacks and unauthorized access.

Cybersecurity risks are a big deal these days. Things like cyberattacks can hit hard, where hackers try to break in and steal stuff or mess things up. It seems like unauthorized access is another worry, maybe someone getting in without permission through weak passwords or something.

- Assess risks of data loss due to human error or technical failures.

There's data loss, which can come from human error, like people accidentally deleting files or clicking bad links. Technical failures add to it too, you know, when systems crash and data just vanishes. It feels like that's more common than we expect, especially if backups aren't done right.

- Consider operational threats such as server outages or power interruptions. Operational threats keep popping up as

well. Server outages might stop everything for hours, and power interruptions can wipe out work in progress. Some people say those are easier to fix than cyber stuff, but they still cause a lot of hassle.

- Risk analysis using a risk assessment matrix:
 - Calculate risk level for each asset based on likelihood of occurrence and impact: The way to figure out risk on assets starts with looking at how likely something goes wrong and then how bad it could get if it does. Basically, you multiply those two things to get the risk level number.

$$\text{Risk Level} = \text{Likelihood} \times \text{Impact} \text{ [6]}$$

- Classify risks as low, medium, high, or very high to prioritize mitigation, sorting them into groups after that makes it easier to handle, like calling some low if they are not a huge issue, or medium for the middle ones that are okay but not great. High would be for risks that could cause real problems, and very high for the worst cases that stick out. Not everything can get fixed right away, so this helps pick what needs attention first.
- Evaluate the impact of risks on academic and administrative operations:
 - Examine how each threat affects service continuity.
Cyber attacks stand out because they hit fast. If hackers get in, they might lock systems or grab data, and then everything grinds to a halt. Recovery is not quick; I think that is what makes it so bad for keeping services going. It just leaves you down for hours or days sometimes.
 - Identify key vulnerabilities that need urgent attention.
Natural disasters are different, like floods or quakes that wreck the physical setup. Servers get damaged, networks fail, and you have to scramble with backups if you even have them set up right.

It seems unpredictable, which drags out the disruption more than you would want.

Hardware issues pop up too, say a server just crashes out of nowhere. That stops whatever it was running, and without a spare ready, continuity breaks right there. It feels routine in a way but still hits hard when it happens.

Power outages tie into that mess. No electricity means nothing runs, and if backup generators fail, it spreads to other areas quickly. That can knock out multiple services at once.

Vulnerabilities make all this worse, I guess. Weak spots in cybersecurity, like old software full of holes, let attacks slip through easily. Then there is not planning backups properly for disasters or those failures, leaving everything exposed. Single points of failure in hardware are another thing, where one breakdown pulls the whole chain down.

Figuring out priorities gets messy, but overlooking tests for continuity plans probably leaves gaps too. Some places skip that, and it shows when trouble hits.

- Propose preventive measures and data recovery mechanisms:
 - Design a regular data backup plan.
When it comes to setting up a regular data backup plan, the key is finding a balance so it happens without too much effort all the time. I think automatic ones that kick in daily make sense, especially if the data does not change a ton every day [8] .
 - Apply the 3-2-1 backup strategy.
That 3-2-1 rule comes up a lot, and it basically says keep three copies total, spread across two kinds of storage, with one kept somewhere else entirely. It helps if local stuff fails, you have backups to fall back on [5] .
 - Develop a disaster recovery plan to restore systems and data as quickly as

possible. A disaster recovery plan pulls all this together by aiming to get things running again quickly after trouble hits. Disasters could be hardware breaking or some cyber issue, so you define those first. Then the plan spells out restoring from the 3-2-1 copies or switching to the replicated database. Prioritizing the important data comes next, and testing restores every so often helps, with a goal like recovering in four hours or less [8] .

- Data collection and analysis:

- Conduct interviews with IT administrators at the university. chatting with a few IT admins from the university. bringing up some daily operation things. Like how much goes into keeping everything running smoothly.
- Review system logs and operational data. Going through the system logs is a bigger deal. There is just so much data piled up. to sift through it all. I mean patterns do show up in the operational stuff. Slowdowns or errors popping here and there.
- Use qualitative and quantitative analysis to identify risks and evaluate the effectiveness of proposed measures. For the analysis, try pulling in notes from those talks. Mixed them with some simple numbers from the logs. It seems like that points out risks better. You know weak spots in security or something like that. Kind of helps connect the dots.

- Propose an integrated framework model:

- Combine risk analysis with data recovery strategies.

Universities have all this sensitive stuff to manage—student records and research data—but most of them are stuck with small budgets and no real tech support. It just seems like without spotting risks first, planning for recovery does not make much sense.

I mean, figuring out threats could start simple; list out things like phishing or bad passwords that happen a lot. Free

online tools might help check that without costing anything. Then, connect it to backups so if an attack hits, you are not totally lost. For instance, wiping files in a cyber thing, a basic backup could pull things together quick.

- Provide practical recommendations to improve information security risk management in universities facing limited technical and financial resources.

Training people on security basics feels key, staff and students both. Workshops with free videos from NIST or something, though I am not totally sure campuses can squeeze that in every time. Human mistakes cause so many problems anyway. On recovery, cheap cloud options like Google Drive for schools could auto save important files. That keeps you from starting over if trouble comes.

Prioritizing what risks matter most, say financial data over other less urgent ones. Helps spend the limited cash on tools that count, like antivirus with backup built in. Some schools skip the thinking part and just buy tech, which might not help.

Testing backups now and then, maybe every few months with help from IT students or volunteers since resources are low. I might be oversimplifying here, but checking risks ongoing could stop little problems growing. That is the part that stands out, keeping it simple.

VI Asset and Threat Analysis

In this study, the critical information assets of Sana’a University have been identified as essential for the continuity of academic and administrative operations. After identifying these assets, potential threats that could affect them were analyzed to assess risks and determine preventive measures.

A. Critical Information Assets

B. Potential Threats

TABLE I
Critical Information Assets at Sana'a University

Asset	Description	Importance Level
Student Information System	Manages student records and grades	High
Student Database	Stores academic and personal data	Very High
University Servers	Hosts applications and databases	Very High
University Network	Connects various systems and enables communication	High

TABLE II
Risk Assessment for Critical Information Assets

Asset	Threat	Probability	Impact	Risk Level
Student Database	Ransomware Attack	Medium	High	High
Student Information System	Unauthorized Access	Medium	High	High
University Servers	Server Failure	Medium	Medium	Medium
University Network	Power or Network Disruption	Low	Medium	Low
Student Database	Data Deletion (Human Error)	Medium	High	High

VII Risk Assessment

After identifying the critical information assets and potential threats, the next step is to evaluate the risks using the OCTAVE Allegro framework. Each risk is assessed based on its probability of occurrence and potential impact on academic and administrative operations to prioritize mitigation actions.

A. Risk Evaluation Table

TABLE III
Risk Assessment for Critical Information Assets

Asset	Threat	Probability	Impact	Risk Level
Student Database	Ransomware Attack	Medium	High	High
Student Information System	Unauthorized Access	Medium	High	High
University Servers	Server Failure	Medium	Medium	Medium
University Network	Power or Network Disruption	Low	Medium	Low
Student Database	Data Deletion (Human Error)	Medium	High	High

VIII Data Recovery Mechanism

To keep things running smoothly in university admin and classes, you really need a solid way to recover data when stuff goes wrong, like security problems or total system crashes, or even bigger disasters. I mean, without that, everything could just stop.

This part is about suggesting some actual steps to take. They tie back to what we talked about earlier with the risks and the key assets that matter most.

Sometimes people overlook how these failures hit the daily operations, but yeah, the recovery setup has to cover that impact.

A. Periodic Backup

Doing backups regularly means you do a full one every week for all the important stuff. Then on most days, you add those smaller incremental ones that just catch the new changes. that way, nothing gets lost if something goes wrong, like if you accidentally delete files or the computer crashes, or even worse with ransomware hitting.

but weekly full ones keep everything covered basically. The daily bits make sure recent work is safe too. Restoring from that should not take forever.

This approach cuts down on how long things are down when problems happen. And yeah, it lowers the chance of losing data for good. Some people might skip the daily part, but it feels important to not risk it.

B. 3-2-1 Backup Strategy

Keeping three copies of important data is what the 3-2-1 backup strategy is all about. It helps with security I guess, so you do not end up losing everything in one go. For universities that handle tons of stuff, this could really cut down on problems from unexpected messes.

You need at least two different types of media for storing those copies. And one of them has to be off site, which seems smart when you think about fires or floods hitting locally. Theft too, that could wipe out what is right there. Spreading things out means primary systems might crash, but something stays safe somewhere.

It feels like a decent plan for keeping data available. Some places probably ignore the off site requirement, and that sounds risky to me. Getting that part handled right is the tough spot, it stands out.

C. Disaster Recovery Plan

A disaster recovery plan is just this setup that tells you how to bring everything back online after a huge problem hits the systems or data. It comes in when there's a big incident and you need to know the next moves right away.

The key stuff in it probably starts with deciding which parts to fix first, like the most critical systems. Then there's getting the databases up and running again and kicking off the university's main operations. Academic side and admin tasks, all that without causing even more chaos. I mean, it feels like that's the core of keeping the place from falling apart completely.

With a DRP in place, the university can deal with tech failures or even security messes in some kind of order. Services keep moving somehow, and they don't lose sight of what they're supposed to do. You know, disruptions get cut back a lot.

IX Results

After applying the research methodology to Sana'a University using the OCTAVE Allegro framework, analyzing critical assets and potential threats, assessing risks, and integrating data recovery mechanisms, the key findings can be summarized as follows:

A. Accurate Identification of Major Risks

The Student Database seems like the biggest deal here from what the analysis pointed out, and it's pretty much right in the middle of everything, so that makes it super exposed to risks.

Then there's the Student Information System too, facing similar stuff.

The threats that stand out are ransomware attacks that could lock things up bad, unauthorized people sneaking in, and even data vanishing from simple human errors.

think those cover the main worries, but maybe there are a few others missed or something.

B. Clear Risk Prioritization

Schools use this probability impact matrix to rank risks based on how bad they could be. It helps sort out the ones that really need attention first. That's the main point of it anyway. Like, university managers can decide where to spend money and time, focusing on the serious threats instead of everything at once. Prioritizing that way makes sense, it keeps things from getting wasted on minor issues.

C. Impact of Data Recovery Strategies

Using periodic backups and that 3-2-1 approach, plus getting a disaster recovery plan in place, helped a ton with keeping security problems from wrecking university stuff like classes or office tasks. It cut down the mess a lot [7].

I think having backups in different spots is key for data safety, you know, so not everything gets lost if something hits. The setup overall keeps services running even when trouble comes up. Without it, things could drag on way longer, with disruptions piling up, or at least it seems that way to me.

Continuity matters most, I guess, but putting it all together right off the bat is tricky sometimes.

D. Enhanced Incident Response Capability

From the results, it looks like putting together risk management and data recovery approaches really helps out the university when dealing with incidents. Like, after a security issue or some technical glitch hits, they can respond faster and get things back to normal more smoothly. Not

totally sure if it's always that straightforward, but combining those strategies seems to boost their overall ability in those situations [6]. That part stands out, anyway.

E. Lessons for Other Institutions

Think the framework they came up with could probably work at other universities too, especially ones that do not have a lot of money or resources. You would just need to tweak it a little for whatever assets are there or the kinds of threats that pop up in those places. It seems adaptable, at least from what I see, and maybe even something you could copy over to similar spots without too much trouble. That part stands out, how it is not stuck to just one setup [5].

F. Summary

Using the OCTAVE Allegro framework with the data recovery plan made it easier to pick out the big risks. We got right to the important ones without wasting time on smaller stuff.

That approach let us figure out what needed fixing first for mitigation. It really helped avoid a lot of those operation headaches that could pop up.

Overall, blending risk management into backups and disaster recovery seems to make the university way more solid against cyber stuff or crashes. Classes and admin tasks keep moving along, not too interrupted or anything.

I think resilience improves a bunch from this.

X Conclusion

Universities without much funding or tech face big problems with information security. All those risks from cyber stuff just hang over them. The study looked at using this thing called OCTAVE Allegro, and they mixed it with ways to back up data if something crashes. It feels practical because you can't fix everything with cash when there's none to spend.

Student databases stand out as super vulnerable. Like, attacks could come in, or someone could sneak access without permission, or data could just get lost by accident. Those systems seem wide open to it all. OCTAVE Allegro is good for spotting the key risks first off. You figure

out which ones are the worst and prioritize them somehow. That way it turns into real steps, not just vague talk about dangers.

On the backup side, they talk about the 3-2-1 rule for copies of everything important. Plus disaster recovery plans to keep classes running and admin stuff from stopping dead. Without that, the whole place could shut down quickly from one failure. It seems like tying the risk checks from OCTAVE into these recovery ideas makes the university tougher against cyber hits or breakdowns.

That link between assessing risks and actually preparing to bounce back is important, but maybe some overlook it. For low-budget spots, clear policies on security risks make sense. Set up those backup procedures solidly. Use a framework like OCTAVE to go through it all structured. Especially when money is tight, being systematic feels key.

Looking forward, this setup could work for other kinds of schools, I suppose. Or maybe automate watching for risks and handling them faster. Not totally sure if that's easy to pull off, though. It might save time if it does.

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